Exercise 1.6

1. **Fill in the blanks:**

   (i) $(-1)^{\text{even integer}}$ is _____.

   (ii) For $a \neq 0$, a^0 is _____.

   (iii) $4^{-3} \times 5^{-3} =$ _____.

   (iv) $(-2)^{-7} =$ _____.

   (v) $\left(-\frac{1}{3}\right)^{-5} =$ _____.

2. **Say True or False:**

   (i) If $8^x = \frac{1}{64}$, the value of x is -2.

   (ii) The simplified form of $(256)^{\frac{-1}{4}} \times 4^2$ is $\frac{1}{4}$.

   (iii) Using the power rule, $\left(3^7\right)^{-2} = 3^5$.

   (iv) The standard form of 2×10^{-4} is 0.0002.

   (v) The scientific form of 123.456 is 1.23456×10^{-2}.

3. Evaluate: (i) $\left(\frac{1}{2}\right)^3$ (ii) $\left(\frac{1}{2}\right)^{-5}$ (iii) $\left(\frac{-5}{6}\right)^{-3}$ (iv) $\left(2^{-5} \times 2^7\right) \div 2^{-2}$ (v) $\left(2^{-1} \times 3^{-1}\right) \div 6^{-2}$

4. Evaluate: (i) $\left(\frac{2}{5}\right)^4 \times \left(\frac{5}{2}\right)^{-2}$ (ii) $\left(\frac{4}{5}\right)^{-2} \div \left(\frac{4}{5}\right)^{-3}$ (iii) $2^7 \times \left(\frac{1}{2}\right)^{-3}$

5. Evaluate: (i) $\left(5^0 + 6^{-1}\right) \times 3^2$ (ii) $\left(2^{-1} + 3^{-1}\right) \div 6^{-1}$ (iii) $\left(3^{-1} + 4^{-2} + 5^{-3}\right)^0$

6. Simplify: (i) $(3^2)^3 \times (2 \times 3^5)^{-2} \times (18)^2$ (ii) $\frac{9^2 \times 7^3 \times 2^5}{84^3}$ (iii) $\frac{2^8 \times 2187}{3^5 \times 32}$

7. Solve for x : (i) $\frac{2^{2x-1}}{2^{x+2}} = 4$ (ii) $\frac{5^5 \times 5^{-4} \times 5^x}{5^{12}} = 5^{-5}$

8. Expand using exponents: (i) 6054.321 (ii) 897.14

9. Find the number in standard form for the following expansions:

   (i) $8 \times 10^4 + 7 \times 10^3 + 6 \times 10^2 + 5 \times 10^1 + 2 \times 1 + 4 \times 10^{-2} + 7 \times 10^{-4}$

   (ii) $5 \times 10^3 + 5 \times 10^1 + 5 \times 10^{-1} + 5 \times 10^{-3}$

   (iii) The radius of a hydrogen atom is 2.5×10^{-11} m

10. Write the following numbers in scientific notation:

    (i) 467800000000 (ii) 0.000001972 (iii) 1642.398

    (iv) Earth's volume is about 1,083,000,000,000 cubic kilometres

    (v) If you fill a bucket with dirt, the portion of the whole Earth that is in the bucket will be 0.000000000000000000000000016 kg

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